



A self-sensing and self-actuating metamaterial sandwich structure for the low-frequency vibration mitigation and isolation

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ABSTRACT

Sandwich structures (SSs) are widely deployed in engineering applications, such as for the low-frequency vibration or acoustic mitigation and isolation. However, it is challenging to enhance their low-frequency vibration mitigation and isolation performance while maintaining their multi-functionalities, such as lightweight, high bending stiffness, and high impact absorption ability, etc. To achieve this objective, local resonators are introduced into a hollow-tube core lattice sandwich structure (HLSS) in this paper, which is referred to as the metamaterial sandwich structure (MMSS-1). The proposed MMSS-1 is composed of periodically distributed thin-walled hollow tubes with attached resonators as the core, and two thin cover-plates or face-sheets. The band structures and the corresponding eigenmodes of the unit-cell of the MMSS show that both the Bragg scattering and the local resonance band-gaps coexist in the proposed MMSS-1. The influences of the key geometrical and material parameters of the MMSS-1 on the band-gaps are analyzed in details. Then, the piezoelectric patches are attached to the top and bottom face-plate surfaces of the MMSS-1 and resonators to form the MMSS-2, and they are shunted via an active feedback control to build the MMSS-3. The effects of the displacement feedback control gains on the band-gaps of the MMSS-3 are studied. Based on the active displacement feedback control and differential evolution (DE) optimization algorithm, a self-sensing and self-actuating strategy is developed to lend the self-adaptive capability to the MMSS-3, which is referred to as the self-adaptive MMSS. The vibration or flexural wave propagation in the finite-sized self-adaptive MMSS is numerically simulated to demonstrate its outstanding ability for low-frequency vibration mitigation and isolation. It shows that the proposed self-adaptive MMSS can evolve different control parameters automatically to widen and merge Bragg scattering band-gaps and local resonance band-gaps without human intervention. In addition, the vibration attenuation inside the band-gaps is also enhanced. The proposed self-sensing and self-actuating strategy can improve the vibration mitigation and isolation performance of the HLSSs significantly in the low-frequency range. The proposed self-adaptive MMSS may pave a novel and feasible way for the design and applications of the HLSSs for the low-frequency vibration mitigation and isolation in civil engineering, mechanical engineering, and aerospace engineering etc. Moreover, the proposed self-adaptive strategy via self-sensing and self-actuating can be also applied to other types of the SSs for improving their vibration mitigation and isolation performance, and for the NVH (noise, vibration and harshness) of vehicles, high-end equipment and high-precision instruments.

1. Introduction

Sandwich structures (SSs) are special composite structures composed of two thin but stiff cover-plates or face-sheets and a lightweight core, which have some unique characteristics, such as their high-stiffness-to-

density ratio, outstanding heat resistance, high impact energy absorption and acoustic damping performances, etc. Among many different kinds of SSs, the lattice sandwich structures (LSSs) are widely used in aerospace [1], automobiles [2] and civil engineering [3]. However, with the rapid development of many cutting-edge and high-precision

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equipment and devices, the low-frequency vibration mitigation and isolation performance of such structures becomes more and more important since the low-frequency vibration may disturb their functionalities, weaken their sensitivity, and even cause their failure. Hence, it is still a challenging task to develop novel LSSs with distinct low-frequency vibration mitigation and isolation properties while keeping their peculiar multi-functionalities, such as their lightweight, high bending stiffness, and high impact absorption ability etc.

Among many possibilities, the concept of artificial periodic structures or phononic crystals (PCs) and acoustic/elastic metamaterials (AMs) provides a novel way for the structural vibration mitigation and isolation [4]. Due to the Bragg scattering effect in such particularly architected periodic materials and structures, the elastic wave propagation is prohibited in certain frequency ranges, which are called the Bragg band-gaps (Bragg-BGs). Besides the Bragg scattering mechanism in the conventional PCs, Liu et al. [5] presented a local resonance mechanism to generate band-gaps, which are often referred to as the local resonance band-gaps (resonance-BGs). Many previously reported LSSs involve a periodic core structure to suppress or isolate the low-frequency vibration [6–12]. Recently, some novel LSSs with broadband low-frequency vibration mitigation and isolation properties have been also proposed. For example, one can break the symmetry of the pyramidal-core lattice sandwich structure via chessboard design and gradient design to widen the band-gaps [13], and Li et al. [14] added the resonators into the pyramidal-core lattice sandwich structure to realize the broadband low-frequency vibration mitigation and isolation.

In recent years, the piezoelectric materials have been widely used to fabricate tunable PCs and AMs [15–17]. For example, the piezoelectric lead zirconate titanate (PZT) patches can be bonded onto the top and bottom surfaces of the LSSs. Ouisse et al. [18] utilized the pyramidal lattice auxetic structure with the shunted PZT patches to control the behavior of flexural waves. Xiao et al. [19] used the shunted PZT patches to improve the low-frequency vibration properties of the carbon fiber reinforced lattice sandwich structure via the resonance circuits. Chen et al. [20] used the shunted PZT patches to enhance the vibration mitigation and isolation properties of the lattice sandwich structure. However, the tuning effects of the PZT patches on the band-gaps of the LSSs are still rather limited.

To enhance the tuning effects of the PZT patches, an adaptive control strategy has been proposed [21]. Moreover, the PZT patches with active feedback control can also be used to build a self-sensing and self-actuating metamaterial structure [22] with special control algorithms. The proposed self-sensing and self-actuating metamaterial structure has the self-adaptive ability and it can adjust itself to various external excitations. Based on this concept, a self-adaptive cloak [23], metasurfaces [24,25], programmable metasurfaces [26], and angular dispersion elimination metasurfaces [27] have been fabricated. These previously reported self-adaptive metamaterials are based on machine learning. As for the acoustic or elastic metamaterials, the population-based algorithms are more widely used [28,29]. Our previous study [30] realized a self-sensing and self-actuating metamaterial plate by bonding the PZT patches onto the plate surfaces, and the PZT patches were shunted via an active feedback control. Then the control parameters can be generated automatically by the metamaterial plate itself via the particle swarm optimization (PSO) algorithm appropriate to different stimulations. Accordingly, the LSSs can be also converted into the self-sensing and self-actuating metamaterial plate structures, and their low-frequency vibration mitigation and isolation properties are expected to be significantly improved.

The hollow-tube core lattice sandwich structure (HLSS) is one of the simplest forms of the LSSs. The HLSS has a superior lightweight with high impact absorption ability [31,32]. Even a small adjustment may largely enhance the vibration mitigation and isolation property of the HLSS. For example, Li et al. [33] and Wang et al. [34] demonstrated that the vibration mitigation and isolation property of the HLSS can be significantly improved by adding a resonator/plate. Qian et al. [35] used

a high-density material embedded into soft materials to build the hollow-tube core, and then the broadband local resonance band-gaps are generated. Cai et al. [36] used the hollow-tube core as a resonator in the LSS to improve the low-frequency mitigation and isolation property.

In this paper, a self-sensing and self-actuating (self-adaptive) metamaterial sandwich structure, which is referred to as the self-adaptive MMSS, is proposed to enhance the low-frequency vibration mitigation and isolation property of the conventional HLSS. For the comparison purpose, three other related MMSSs are also considered, which are summarized in Table 1, where their essential components are also given. The proposed MMSS-1 consists of two thin cover-plates or face-sheets, periodically distributed hollow tubes and resonators as shown in Fig. 1a. Hence, the vibration mitigation and isolation performance is improved by the local resonance mechanism induced by local resonators. The effects of the geometrical and material parameters of the resonators on the band-gaps are analyzed. In the MMSS-2, the PZT patches are additionally bonded onto the surfaces of the MMSS-1 and the resonators, among which some PZT patches act as sensors while the others act as actuators, as shown in Fig. 1a. In the MMSS-3, the PZT patches of the MMSS-2 are shunted via an active feedback control. In the self-adaptive MMSS, the differential evolution (DE) algorithm [37] is adopted in the feedback control strategy, which lends the self-adaptive capability to the MMSS-3. In comparison to many previous investigations on the HLSSs, the present work has the following novel objectives:

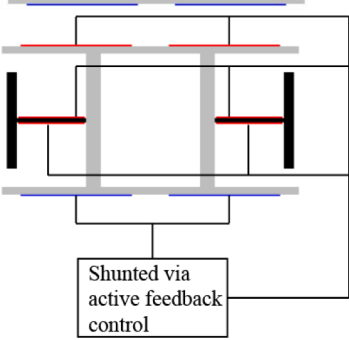
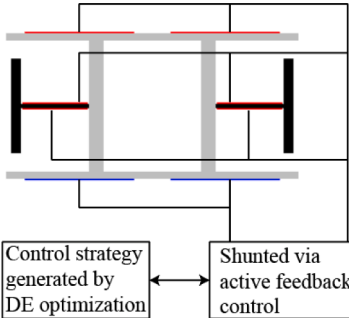
- **Band-gap merging:** The self-adaptive MMSS exhibit both Bragg scattering and local resonance band-gaps, which should be merged to a wider band-gap (see Fig. 1b). Such a band-gap merging via self-adaptive metamaterials is very rarely reported in the previous studies [38,39].
- **Mitigation enhancing:** The proposed self-adaptive MMSS should enhance the vibration mitigation or attenuation inside and outside the band-gaps.
- **Self-adaptivity:** The proposed self-adaptive MMSS should automatically select the appropriate control parameters according to the different external excitations without human intervention.

The rest of the paper is organized as follows. The proposed metamaterial sandwich structure (MMSS-1) is presented in Section 2, where the procedure for computing the band structures is described, and the effects of the key geometrical and material parameters of the MMSS-1 on the band-gaps are investigated. Section 3 presents an active displacement feedback control method for the MMSS-3 by shunting the piezoelectric patches on the surfaces of the MMSS-2. The influences of the displacement feedback control gain on the band-gaps are analyzed in details. The self-adaptive MMSS with a self-sensing and self-actuating capability is presented in Section 4. The self-adaptivity is guided by the differential evolution (DE) optimization algorithm to match the external excitation automatically. Representative examples demonstrate the high efficiency and performance of the self-adaptive MMSS for the vibration mitigation and isolation. Finally, some conclusions from this study are given in Section 5.

2. Metamaterial sandwich structure (MMSS-1)

Fig. 2a shows a schematic sketch for a unit-cell of the MMSS-1 assembled by a hollow-tube core lattice sandwich structure (HLSS) marked by gray and a resonator marked by cyan. Its first irreducible Brillouin zone is shown in Fig. 2c. The thin-walled hollow tubes are periodically distributed in both *x*- and *y*- directions and form the lightweight core embraced by two thin cover-plates or face-sheets. The hollow-tube core, the top and the bottom cover-plates could be composed of different materials. The top and bottom cover-plates could also have different thicknesses. The HLSS has a superior lightweight with a high impact-absorption ability [31,32] and out-of-plane stiffness.

Table 1
Comparison of the unit-cells of the present MMSSs.

MM models	Schematic diagrams	Component
MMSS-1		Hollow-tube core lattice sandwich structure (HLSS) with a resonator
MMSS-2		PZT patches are bonded to the surfaces of the MMSS-1 and the resonator
MMSS-3		PZT patches of the MMSS-2 are shunted via active feedback control
Self-adaptive MMSS		Control strategy of the MMSS-3 is evaluated and generated by the differential evolution (DE) optimization algorithm

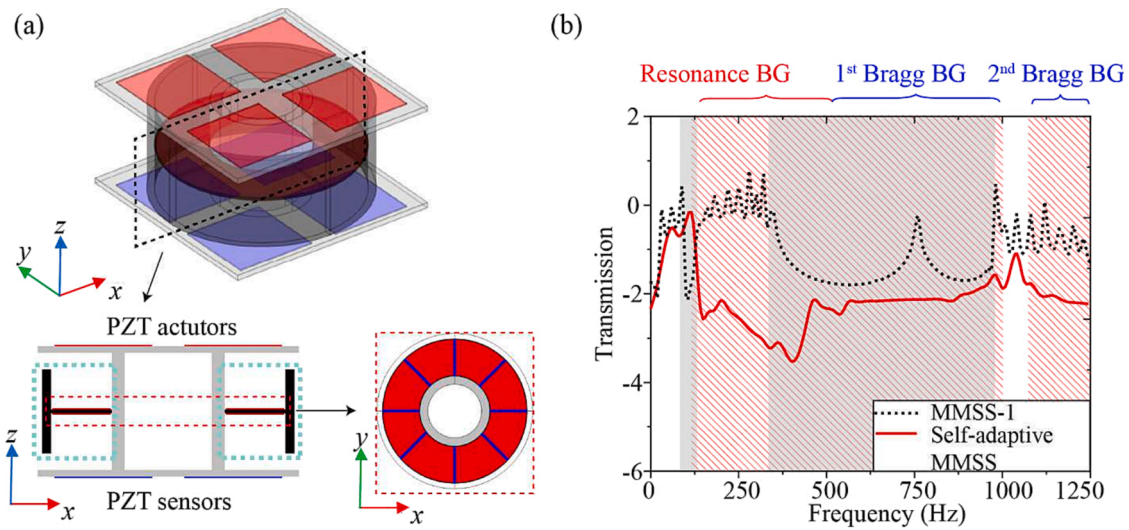


Fig. 1. (a) The schematic diagram for the unit-cell of the self-adaptive MMSS. The blue PZT patches act as sensors, and the red PZT patches as actuators. The part in the green box is the resonator. (b) The band-gaps of the MMSS-1 and the self-adaptive MMSS. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

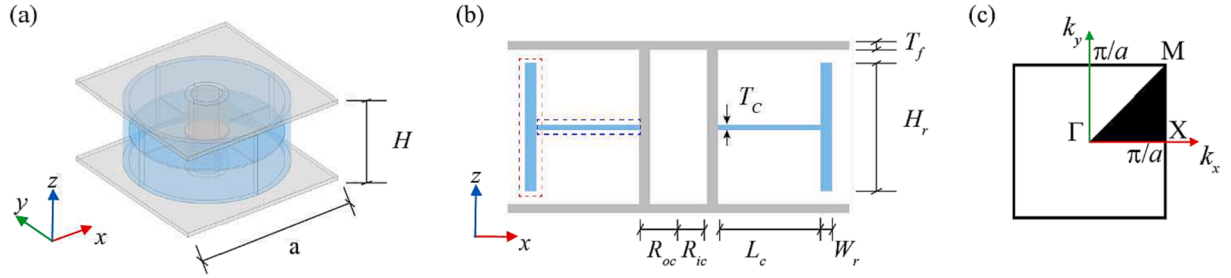


Fig. 2. (a) The schematic diagram for the unit-cell of the MMSS-1. (b) The front view of the unit-cell of the MMSS-1. The part in the blue rectangle is the connector of the resonator (circular ring-plate), and the part in the red rectangle is the additional mass block of the resonator (circular ring-beam). (c) The first irreducible Brillouin zone. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

To enhance the vibration mitigation and isolation properties of the conventional HLSS, we add a resonator (the cyan part) to each unit-cell of the MMSS-1, as shown in Fig. 2a and 2b. The resonator is composed of a thin circular ring-plate as connector (in the blue rectangle) and a heavy circular ring-beam as additional mass (in the red rectangle). For the following calculations in this section, we set the lattice constant as $a = 0.2$ m, the height of the MMSS-1 as $H = a/2$, the outer tube radius as $R_{oc} = 0.7a/3$, the inner tube radius as $R_{ic} = 0.5a/3$, and the thickness of the face-sheets as $T_f = 0.1a/2$. For the resonator, the thickness of the connector is $T_c = 0.002$ m, the radius of the connector is $L_c = 1.8a/3$, the height of the circular ring-beam is $H_r = 0.75a/2$, and the width of the circular ring-beam is $W_r = 0.2a/3$. The HLSS and the connector of the resonator are made of aluminum with the mass density being 2700 kg/m^3 , the Young's modulus being 70 GPa , and the Poisson's ratio being 0.35 . While the circular ring-beam is made by lead with the mass density being 11600 kg/m^3 , the Young's modulus being 40.8 GPa , and the Poisson's ratio being 0.369 .

In the absence of body forces, the governing equations for the elastic

wave propagation in the MMSS-1 can be written as

$$\nabla \cdot [C : \nabla u(r, t)] = \rho \frac{\partial^2 u(r, t)}{\partial t^2}, \quad (1)$$

in which $\nabla = (\partial/\partial x, \partial/\partial y, \partial/\partial z)$ is the differential operator; $u(r, t)$ is the displacement vector; $r = (x, y, z)$ is the position vector, t is the time; C and ρ are the elasticity tensor and mass density, respectively. For the time-harmonic wave motion with the angular frequency ω , Eq. (1) can be rewritten as

$$\nabla \cdot [C : \nabla u(r)] = -\rho \omega^2 u(r), \quad (2)$$

where the common time-harmonic factor $e^{i\omega t}$ is omitted throughout the paper for the sake of brevity.

By applying the finite element method (FEM) and the Bloch periodic boundary conditions to the unit-cell of the MMSS-1, one can obtain the following discrete eigenvalue equation

$$(K - \omega^2 M)u = 0, \quad (3)$$

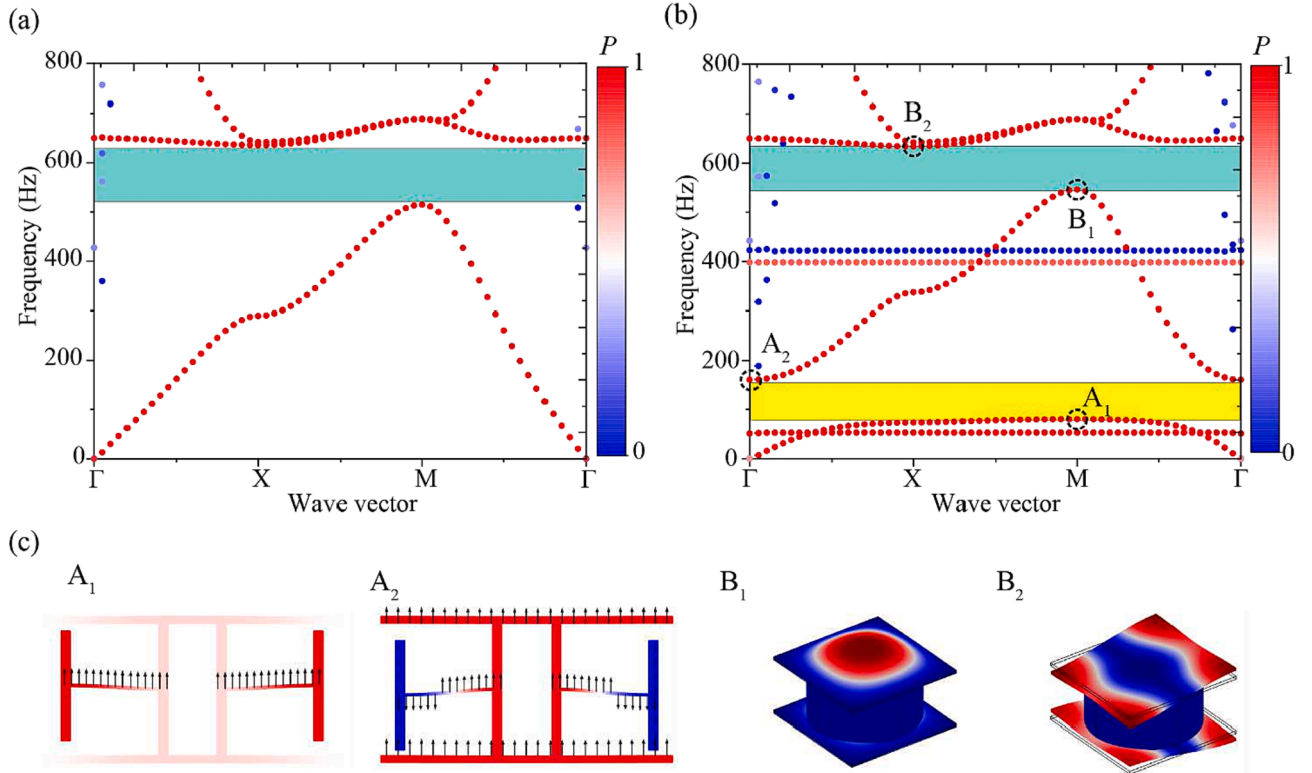


Fig. 3. The band structures of (a) the HLSS, and (b) the MMSS-1. The corresponding edge modes of the MMSS-1 are shown in (c). The shaded areas in (a) and (b) represent the band-gaps. The green shaded areas are the Bragg-BGs, while the yellow shaded area is the resonance-BG. The blue arrows illustrate the displacement directions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where $\mathbf{K}$ and $\mathbf{M}$ are the stiffness and mass matrices, respectively, and $\mathbf{u}$ is the vector of the nodal displacements. In this study, Eq. (3) is solved by the commercial FEM software COMSOL Multiphysics. When the resonator is introduced into the unit-cell, the local resonance modes are also generated. The following polarization index is used to distinguish the flexural wave modes from the bulk wave modes

$$P = \frac{\iiint |w| dV}{\iiint \sqrt{|u|^2 + |v|^2 + |w|^2} dV}, \quad (4)$$

where u , v and w represent the displacement components in the x -, y -, and z -directions, respectively.

The band structures of the HLSS and MMSS-1 are shown in Fig. 3a and 3b, respectively, where the color scales represent the polarization index P . It can be seen that the resonator induces a new band-gap (50–90 Hz) compared to the HLSS. The corresponding eigenmodes at the edges of the band-gaps are shown in Fig. 3c. For the modes A_1 and A_2 , only the core of the MMSS-1 or the resonator vibrates, which means that the first band-gap in Fig. 3b is a resonance-BG, which is not present in the HLSS as shown in Fig. 3a. By contrast, the modes B_1 and B_2 are the typical flexural wave modes triggered by the so-called trampoline effect [40]. Therefore, the second band-gap is the 1st Bragg-BG. It should be pointed out here that the resonator almost has no influence on the 1st Bragg-BG (524–634 Hz). In the following part, the influences of the key geometrical and material parameters of the MMSS-1 on the band-gaps will be investigated.

In Fig. 4a, the influences of the thickness of the face-sheets T_f on the band-gaps of the MMSS-1 are shown. It is found that, with the increase of T_f , the 1st Bragg-BG moves to higher frequency, while the resonance-BG shrinks. The reason is that the mass and the stiffness of the MMSS-1 both increase with T_f . This conclusion was also drawn by Ma. et al. [8] and Yu et al. [9], who studied the elastic wave propagation in the HLSS. However, the resonance-BG only shrinks a little bit with the increase of T_f .

In Fig. 4b, the effects of the inner tube radius R_{ic} for fixed outer tube radius R_{oc} on the band-gaps are also presented. The inner tube radius R_{ic} almost has no effect on the resonance-BG and the upper edge of the 1st Bragg-BG. But it is a different situation for the lower edge of the 1st Bragg-BG, it moves to higher frequency with the increase of R_{ic} . The reasons are the reduced mass of the core and the enhanced connection between the core and the face-sheets with the increase of R_{ic} for fixed R_{oc} . Therefore, the 1st Bragg-BG shrinks.

The effects of the radius parameter L_c and the thickness T_c of the connector on the resonance-BG and the 1st Bragg-BG are shown in Fig. 5.

It can be seen that the resonance-BG disappears when L_c is smaller than $0.2a/3$, as shown in Fig. 5a. It means that when the connector is shorter than $0.2a/3$, the resonator acts only as an additional mass to the core of the MMSS-1. When L_c is larger than $0.2a/3$, the resonance-BG appears as shown in Fig. 3b. Moreover, the lower and upper edges of the band-gap when $L_c = 0.2a/3$ correspond to the modes A_1 and B_2 shown in Fig. 3c, respectively. In contrast, the thickness T_c shows a totally different effect on the band-gaps. Since the effective stiffness of the connector increases with T_c , the resonance-BG becomes wider and moves to a higher frequency range. Then the 1st Bragg-BG is in this case submerged by the resonance-BG.

The effects of the height H_r and the width W_r of the circular ring-beam (mass block) on the resonance-BG and the 1st Bragg-BG are presented in Fig. 6. It can be seen from Fig. 6 that the resonance-BG shifts to a lower frequency range when H_r and W_r increase, while H_r and W_r have only slight effects on the 1st Bragg-BG. The resonance-BG disappears when H_r or W_r is too small, as shown in Fig. 6a, which also confirms that the resonance-BG is induced by the local resonance of the resonators.

The influences of the Young's modulus and the mass density of the resonator on the band-gaps are shown in Fig. 7. Here, we vary either the Young's modulus and the mass density of the circular ring-beam (mass block) or the thin circular ring-plate (connector) while keep other material parameters unchanged to evaluate the effects of the material parameters and their combinations on the band-gaps. It can be seen that with the increase of the Young's modulus of the connector E_{ct} the resonance-BG moves to a higher frequency range and it becomes wider, as shown in Fig. 7a. However, the 1st Bragg-BG barely changes as long as E_{ct} is small, but it becomes narrower with increasing E_{ct} until it is submerged by the resonance-BG when E_{ct} is sufficiently large. Fig. 7b shows that the mass density of the connector ρ_{ct} has a weak influence on the resonance-BG, and also only a slight effect on the 1st Bragg-BG in the range of $\rho_{ct}/\rho \leq 10$ while a significant influence on the 1st Bragg-BG when $\rho_{ct}/\rho > 10$. When the connector's mass density is high enough, the connector can be considered only as an additional mass to the tube core, which moves the 1st Bragg-BG to a lower frequency range as ρ_{ct} increases. It is shown in Fig. 7c that the Young's modulus of the additional mass block E_{ad} almost has no effects on both band-gaps. However, the mass density of the additional mass block ρ_{ad} has a remarkable influence on the resonance-BG while only a weak effect on the 1st Bragg-BG, as shown in Fig. 7d.

In summary, the geometrical and material parameters of the resonator determine its stiffness and mass and hence its resonance frequencies, which are vital to the generation of the local resonance band-

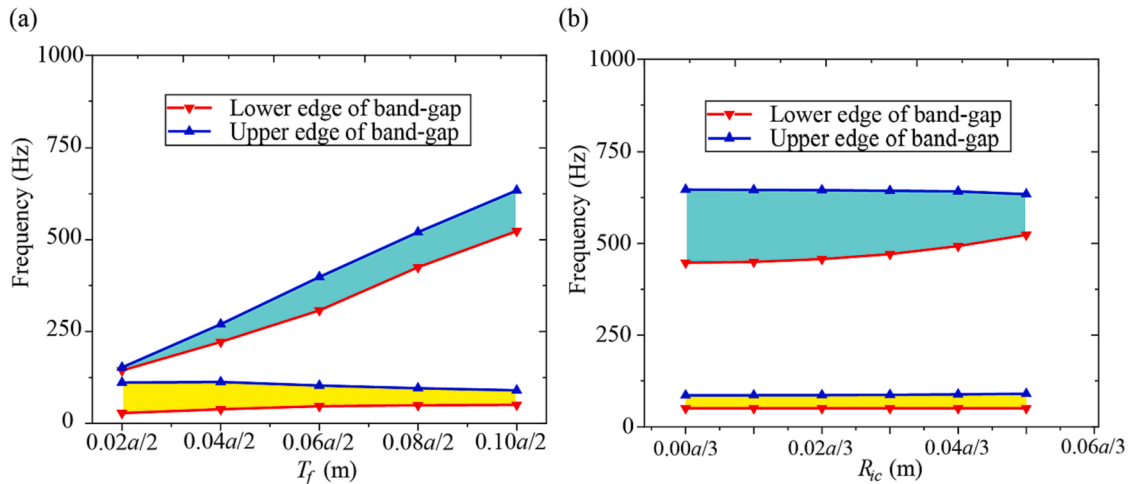


Fig. 4. The influences of (a) the thickness of the face-sheets T_f , and (b) the inner tube radius R_{ic} on the band-gaps. The green shaded areas are the Bragg-BGs, while the yellow shaded areas are the resonance-BGs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

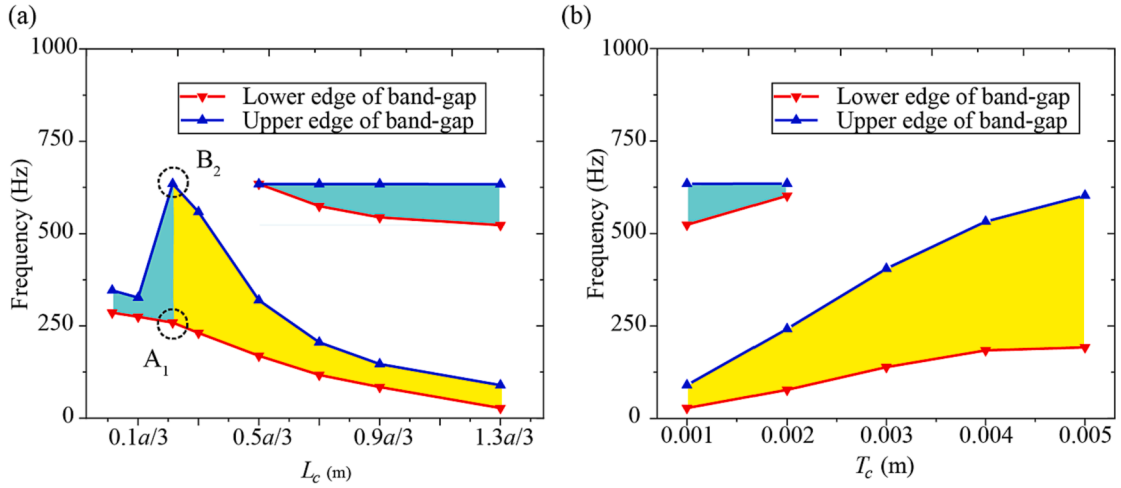


Fig. 5. The effects of (a) the radius parameter L_c , and (b) the thickness T_c of the connector on the band-gaps. The green shaded areas are the Bragg-BGs, while the yellow shaded areas are the resonance-BGs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

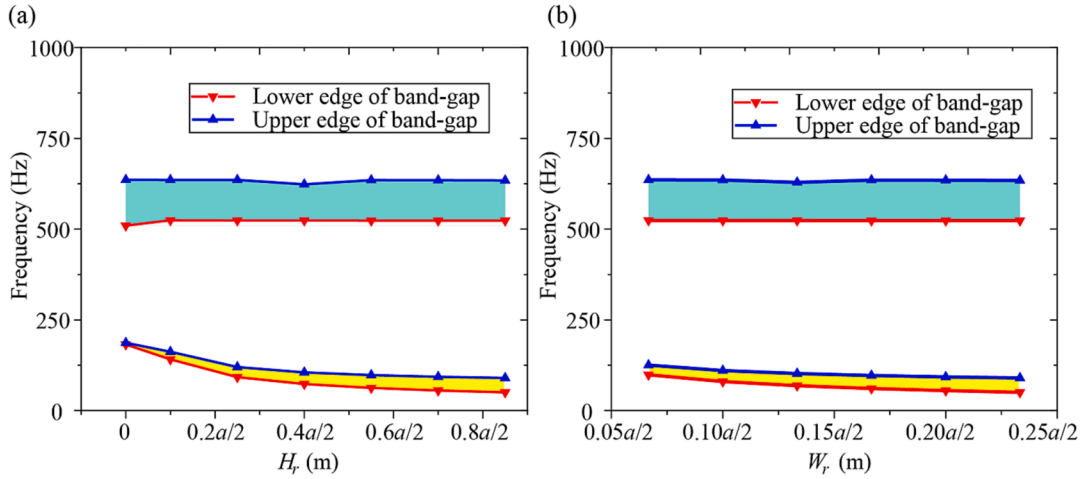


Fig. 6. The effects of (a) the height H_r , and (b) the width W_r of the ring-beam (mass block) on the band-gaps. The green shaded areas are the Bragg-BGs, while the yellow shaded areas are the resonance-BGs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

gaps. On the other hand, the key parameters that affect the Bragg scattering band-gaps are the thickness of the face-sheets T_c , the inner tube radius R_{ic} (for fixed R_{oc}) and the thickness of the hollow-tube core $R_{oc} - R_{ic}$, as well as the geometrical and material parameters of the resonator. From Figs. 4-7, we can conclude that the combination of the small values of T_f , T_c and W_r with the large values of H_r , R_{oc} , R_{ic} and L_c could simultaneously lead to a lowest resonance-BG edge and a highest 1st Bragg-BG edge while keeping the high stiffness-to-density ratio of the MMSS-1.

3. Active feedback control of the metamaterial sandwich structure

In this section, the PZT patches are bonded onto the surfaces of the MMSS-1 and the resonators to form the MMSS-2, as shown in Fig. 8. The displacement feedback control method is adopted in this study, which is defined as.

$$U_{zpa} = -g_d U_{zps}, \quad (5)$$

where g_d is the displacement feedback control gain, U_{zpa} and U_{zps} are the out-of-plane voltage of the piezoelectric actuators and sensors, respectively. For simplicity, the PZT patches bonded to the connectors are considered as a thin circular piezoelectric ring-plate as shown in Fig. 8b

and 8d, a minor part of which acts as sensor while the other major part acts as actuator. To simplify the analysis, the minor sensor part is neglected, and the total circular piezoelectric ring-plate is considered as an actuator in the following calculations. Then, different displacement feedback control gains (g_{d1} and g_{d2}) are applied to the different PZT actuators, as shown in Fig. 8b.

The governing equation for the base plate with the shunted PZT patches via the displacement feedback control as shown in Fig. 8c can be derived by the Hamilton principle and the Kirchhoff thin-plate theory [30]. The governing equation is given by

$$\bar{m}_p \frac{\partial^2 w}{\partial t^2} + \bar{D}_p \nabla^2 \nabla^2 w = 0, \quad (6)$$

where

$$\bar{m}_p = \rho_b h_b + \rho_{pa} h_{pa} + \rho_{ps} h_{ps}, \quad (7)$$

$$\bar{D}_p = \frac{E_b h_b}{12(1-\nu_b)} + D_{ps} + D_{pa} + \frac{e_{31}}{4\epsilon_{33} h_{pa}} \left[\left(\frac{h_b}{2} + h_{pa} \right)^2 - \left(\frac{h_b}{2} \right)^2 \right]^2 (2 + g_d), \quad (8)$$

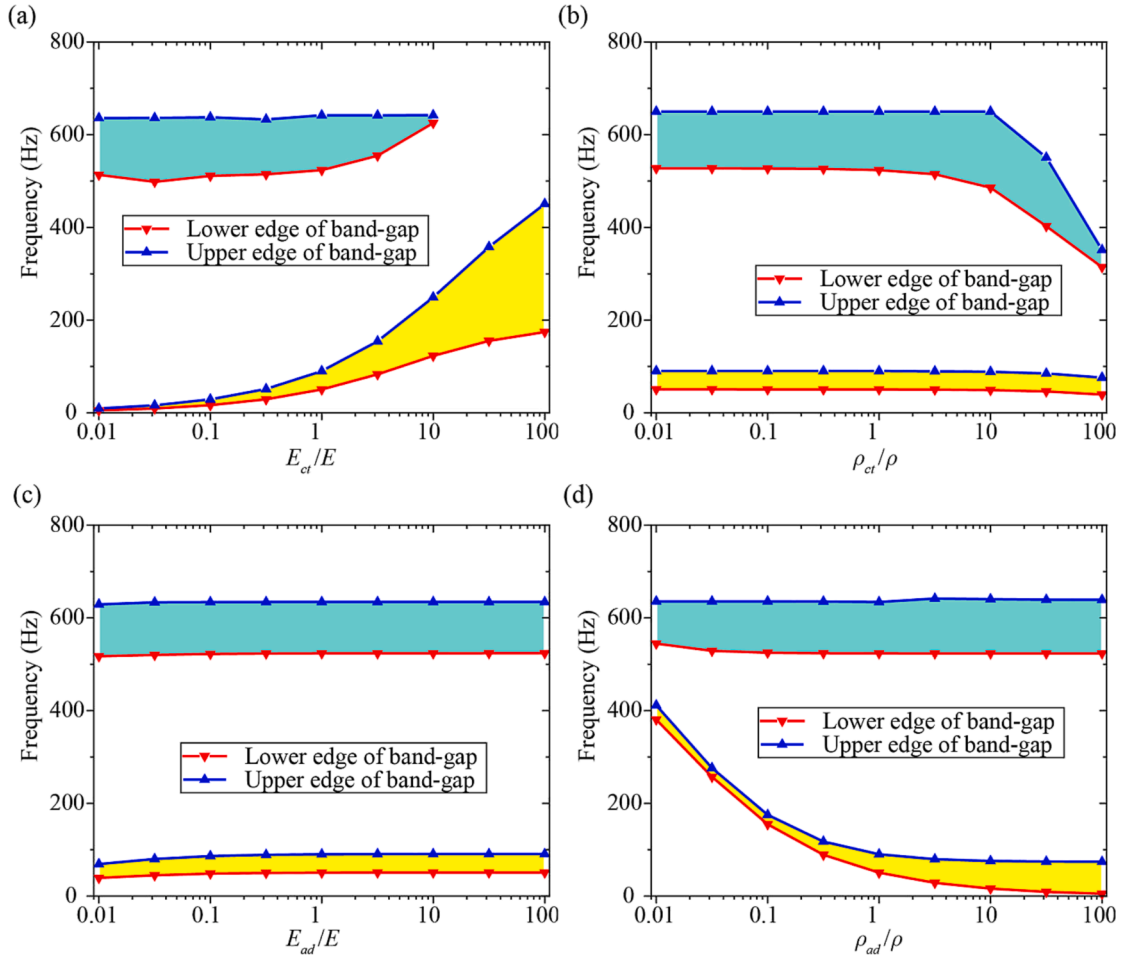


Fig. 7. The effects of (a) and (c) the Young's modulus, and (b) and (d) the mass density of the resonator on the band-gaps. Here, $E_{ct}(E_{ad})$ and $\rho_{ct}(\rho_{ad})$ represent the Young's modulus and the mass density of the connector (the additional mass block), respectively. The logarithmic scale is used for the horizontal coordinates. The green shaded areas are the Bragg-BGs, while the yellow shaded areas are the resonance-BGs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$D_{pa} = \frac{E_{pa}}{3(1-\nu_{pa}^2)} \left[\left(\frac{h_b}{2} + h_{pa} \right)^3 - \left(\frac{h_b}{2} \right)^3 \right], \quad (9)$$

$$D_{ps} = \frac{E_{ps}}{3(1-\nu_{ps}^2)} \left[\left(\frac{h_b}{2} + h_{ps} \right)^3 - \left(\frac{h_b}{2} \right)^3 \right], \quad (10)$$

in which w is the out-of-plane displacement, ρ_b and h_b are the mass density and the thickness of the base plate to which the PZT patches are attached, E_b and ν_b are the Young's modulus and the Poisson's ratio of the base plate, $E_{pa} = E_{ps}$ and $\nu_{pa} = \nu_{ps}$ are the Young's modulus and the Poisson's ratio of the PZT sensors and actuators, $h_{ps} = h_{pa}$ is the thickness of the piezoelectric sensors and actuators, e_{31} is the reduced piezoelectric constant, and ϵ_{33} is the reduced dielectric permittivity under constant strain, respectively. Note that all the terms in Eqs. (6)–(10) with the subscript “ps” should be deleted, and $2+g_d$ should be replaced by $1+g_d$ for the plate only with the PZT actuators in Fig. 8b and 8d. For the plate only with the PZT sensors in Fig. 8b, all the terms in Eqs. (6)–(10) with the subscript “pa” should be deleted, and $2+g_d$ should be replaced by 1.

Based on our previous research [30], the edges of the widened band-gaps for the self-sensing and self-actuating metamaterial structure can be determined by the active displacement feedback control. In order to achieve the lowest and the highest band-gap edges at the same time, the geometrical parameters are chosen based on the results of the last

section. Then the geometrical parameters are selected as $a = 0.2$ m, $H = a/4$, $T_f = 0.1a/2$, $R_{oc} = 1.3a/6$, $R_{ic} = a/6$, $T_c = 0.001$ m, $L_c = 0.7a/3$, $H_r = 0.85a/2$ and $W_r = 0.1a/3$ for the MMSS-2 shown in Fig. 8. The length of the quadratic PZT patches a_p on the face-sheets is taken as $0.75a/2$, and the thickness of the PZT patches T_p is 0.001 m. The aluminum is chosen for the MMSS-2, and PZT-5H is used for the sensors and actuators. The material parameters of the PZT-5H can be found in [41].

The band structures of the MMSS-2 are shown in Fig. 9a. It can be seen that the bonded PZT patches shift the resonance-BG and the 1st Bragg-BG to a higher frequency range due to the increase of the stiffness of the face-sheets and the connectors. It should be noted here that the asymmetric and symmetric modes (see Fig. 9b) are folded together at the M point, see Fig. 9a, due to the symmetry of the MMSS-2 in the z -direction. However, once the active displacement feedback control is applied, the symmetry of the MMSS-2 in the z -direction is broken. In this way, the B_1 and B_2 modes will be separated. That will generate a second Bragg scattering band-gap (2nd Bragg-BG, see Fig. 10b).

The effects of the displacement control gains g_{d1} and g_{d2} on the band-gaps of the MMSS-3 are shown in Fig. 10a and 10b, respectively. The resonance-BG increases with increasing g_{d1} as shown in Fig. 10a. Its lower and upper edges also shift to higher frequency with the increase of g_{d1} . However, the upper edge of the 1st Bragg-BG is not sensitive to g_{d1} , which is related to the effect of the Young's modulus of the connector shown in Fig. 7a. In contrast, the lower edge of the 1st Bragg-BG moves to higher frequency as g_{d1} increases. Fig. 10b shows that g_{d2} has nearly

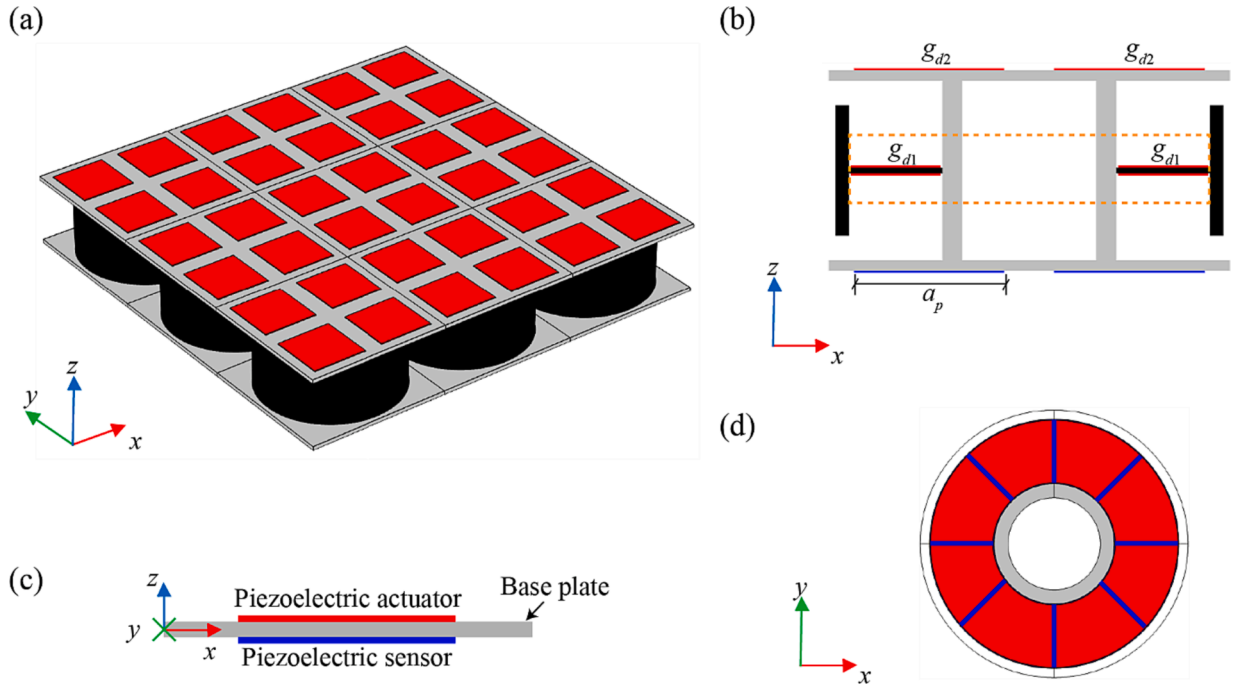


Fig. 8. (a) The schematic diagram of the MMSS-2. (b) The front view of the unit-cell of the MMSS-2 (b). (c) The simplified model of the shunted PZT patches attached to a base plate. (d) The top view of the resonator marked by dashed orange lines in (b). The PZT patches marked by blue are the sensors, and the PZT patches marked by red are the actuators. The PZT patches are shunted via the active feedback control. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

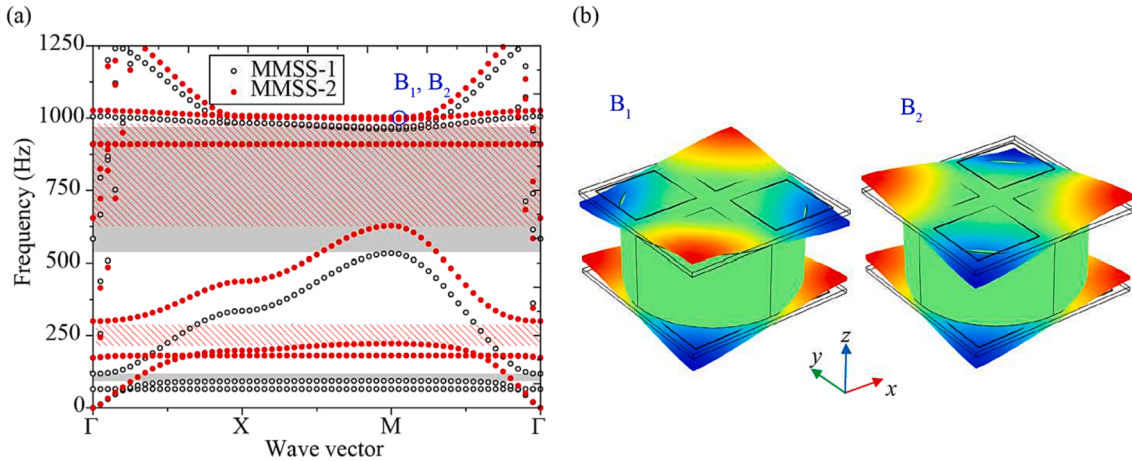


Fig. 9. (a) The band structures of the MMSS-1 and MMSS-2, where the shaded areas denote the band-gaps. (b) The asymmetric and symmetric modes in the z -direction at the upper edge of the 1st Bragg-BG marked by the blue circle in (a). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

no effect on the resonance-BG. When $g_{d2} < 0$, the frequency range of the 1st Bragg-BG increases with the increase of g_{d2} . When $g_{d2} > 0$, only the lower edge of the 1st Bragg-BG increases while the upper edge of the 1st Bragg-BG remains the same. The reason is that the stiffness of the face-sheets will increase with the increase of g_{d2} . In addition, since g_{d2} breaks the symmetry of the MMSS-3 in the z -direction, the symmetric and asymmetric modes in Fig. 9b will be separated. Therefore, the 2nd Bragg-BG is generated, as shown in Fig. 10b. It can be seen that the 2nd Bragg-BG disappears when $-0.5 < g_{d2} < 2.5$. That is because that the symmetry in the z -direction of the MMSS-3 is maintained. When $g_{d2} > 2.5$ or $g_{d2} < -0.5$, the 2nd Bragg-BG appears and shifts to a higher frequency range with the increase of g_{d2} .

The above presented results show that a simple active displacement feedback control strategy cannot widen and move the band-gaps to

lower frequency range at the same time. Hence, we propose to use the self-adaptive scheme via self-sensing and self-actuating to tune the band-gaps by relating the displacement control gains to the frequency. Then we can simply merge the BGs via different frequency-dependent control gains. Accordingly, the target band structures of the self-sensing and self-actuating MMSS (self-adaptive MMSS) are presented in Fig. 11. Although the displacement control gain g_{d1} or g_{d2} to achieve the target band structures in Fig. 11 is unknown in certain frequency ranges, it can still be expected that the self-adaptive MMSS should have the following two essential features. Firstly, the self-adaptive MMSS can widen and move the resonance-BG, the 1st Bragg-BG, and the 2nd Bragg-BG to lower frequencies simultaneously. Secondly, the self-adaptive MMSS can merge the resonance-BG and the 1st Bragg-BG.

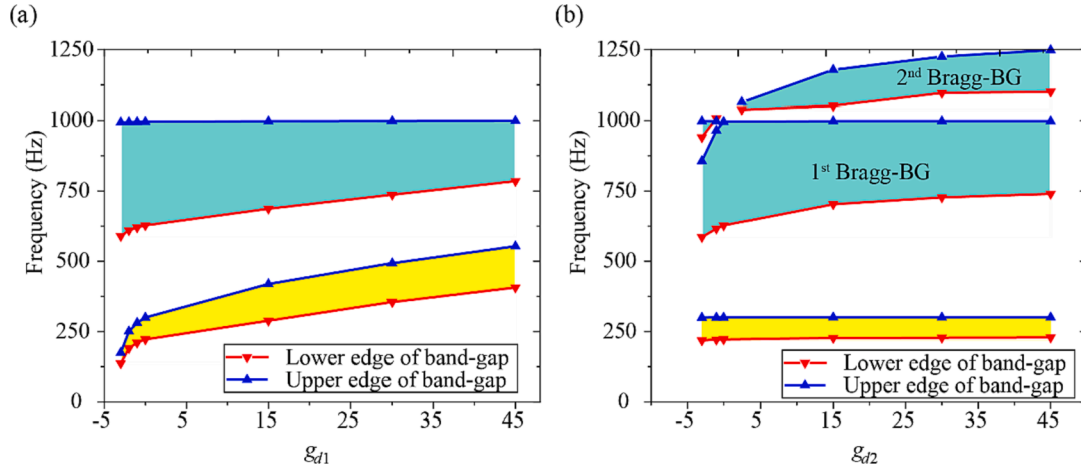


Fig. 10. The effects of the displacement feedback control gains (a) g_{d1} and (b) g_{d2} on the band-gaps of the MMSS-3. The green shaded areas are the Bragg-BGs, while the yellow shaded areas are the resonance-BGs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

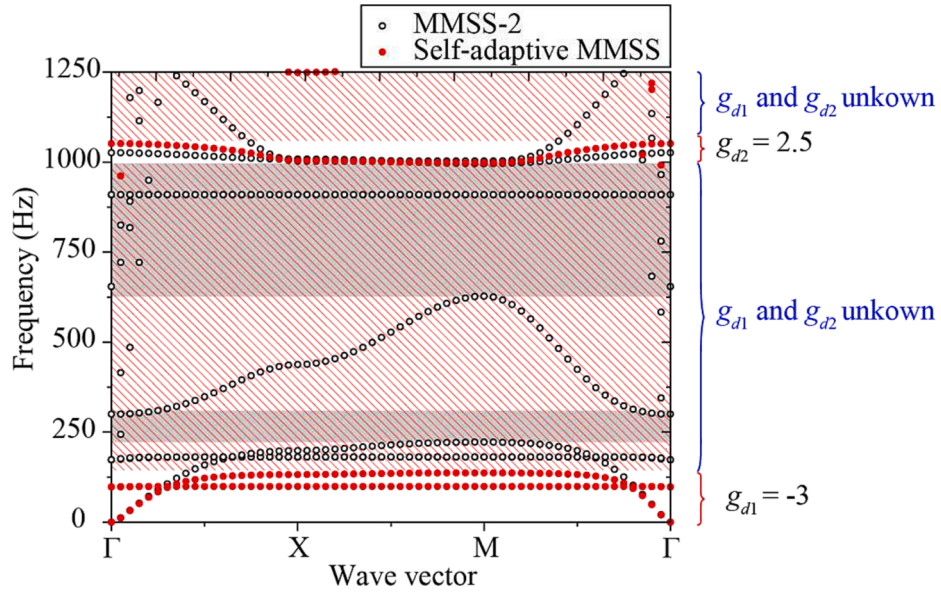


Fig. 11. The target band structures of the self-sensing and self-actuating MMSS (self-adaptive MMSS). The band-gaps of the MMSS-2 by gray, while the target band-gaps of the self-adaptive MMSS are shaded by red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Self-sensing and self-actuating MMSS (self-adaptive MMSS)

In order to investigate the vibration or flexural wave propagation characteristics of the self-adaptive MMSS numerically, we analyze the

dynamic frequency responses or transmission spectra of a finite-sized self-adaptive MMSS as shown in Fig. 12. To show the wide applicability of the proposed structure, different boundary conditions and stimulations are considered. A line source (colored by red) is applied at

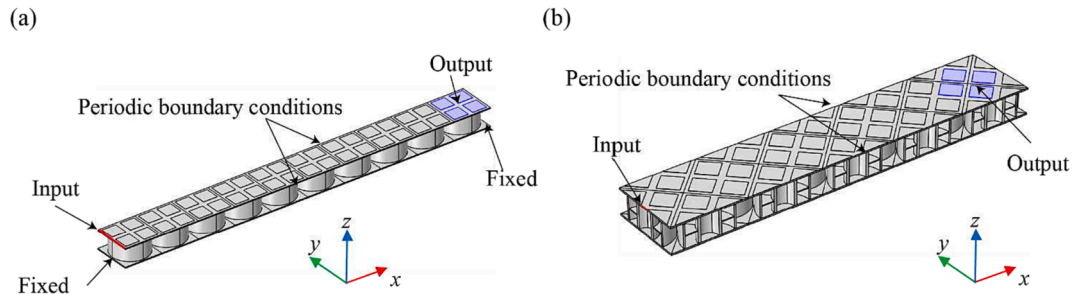


Fig. 12. Schematic diagrams for computing the transmission spectra of the self-adaptive MMSS with the bonded PZT patches in the (a) GX direction, and (b) GM direction.

the upper face-sheet of the self-adaptive MMSS. Ten unit-cells are used in this MMSS to achieve both the computational efficiency and the numerical accuracy. The bottom edges of the self-adaptive MMSS in the x -direction are fixed for computing the transmission spectra in the ΓX direction (see Fig. 12a), while the free boundary conditions are assumed for the faces of the self-adaptive MMSS in the x -direction for computing the transmission spectra in the ΓM direction (see Fig. 12b). The periodic boundary conditions are applied in the y -direction. Then we define the following transmission coefficient.

$$T = \log_{10} \frac{\frac{1}{S_{ps}} \iint |w_{out}| dS}{S_{ps} w_{in}}, \quad (11)$$

in which w_{out} is the out-of-plane displacement of the PZT sensors at the output or receiving area in Fig. 12, S_{ps} is the area of the PZT sensors, and $w_{in} = 1$ is the amplitude of the line source input.

The proposed self-adaptive MMSS is a self-sensing and self-actuating structure. Therefore, an efficient algorithm (such as deep-learning, PSO, etc.) must be adopted to communicate between the sensors and actuators for realizing the self-sensing and self-actuating operation. In this paper, we choose the differential evolution (DE) algorithm, which is shown in the dashed frame of the flow chart of the self-adaptive MMSS in Fig. 13a. The optimization problem at hand is defined by.

Minimize: $T(g_{d1}, g_{d2})$,

$$\text{Subject to : } g_{d1} \in [g_d, g_d], \quad g_{d2} \in [g_d, g_d], \quad (12)$$

in which $g_d = -3, g_d = 45$, and T is the transmission coefficient defined in Eq. (11). To obtain an optimal solution, the settings of the main parameters of the DE (population size, scaling factor, and crossover probability) can be found in [37]. The key steps of the DE algorithm for the proposed self-adaptive MMSS can be summarized as follows.

(i) **Initialization:** 20 uniformly distributed populations are

generated, which are denoted as X_i with $i = 1, 2, 3 \dots 20$. Here, X_i represents a vector defined by.

$$X_i = \begin{bmatrix} X_{i,1} \\ X_{i,2} \end{bmatrix} = \begin{bmatrix} g_{d1}(i) \\ g_{d2}(i) \end{bmatrix}, \quad (13)$$

in which the superscript G denotes the generation number, i denotes the population number, and $X_{i,j}$ ($j = 1, 2$) denotes the member in each population. Then the fitness of the population X_i is calculated and denoted by $T(X_i)$.

(ii) **Mutation:** A mutant vector for each population at the generation G is generated as.

$$M_i = X_{r1} + F(X_{r2} - X_{r3}), \quad (14)$$

in which $F = 0.8$ is the scaling factor and the subscripts $r1, r2, r3 \in \{1, 2, \dots, 20\}$ indicate the distributed randomly chosen vectors.

(iii) **Crossover:** After mutation, the new population NX_i is generated by.

$$NX_i = \begin{cases} M_{i,j}, & \text{if } \text{rand}(0, 1) < C_r \\ X_{i,j}, & \text{if } \text{rand}(0, 1) \geq C_r \end{cases}, \quad (15)$$

where $C_r = 0.9$ is the crossover probability.

(iv) **Selection:** In this operation, a comparison is made between the NX_i and X_i according to their fitness. Then the next generation is chosen as.

$$X_i = \begin{cases} NX_i, & \text{if } T(NX_i) > T(X_i) \\ X_i, & \text{if } T(NX_i) \leq T(X_i) \end{cases} \quad (16)$$

(v) **Repeat steps (ii) to (iv)** until a predefined termination criterion is satisfied.

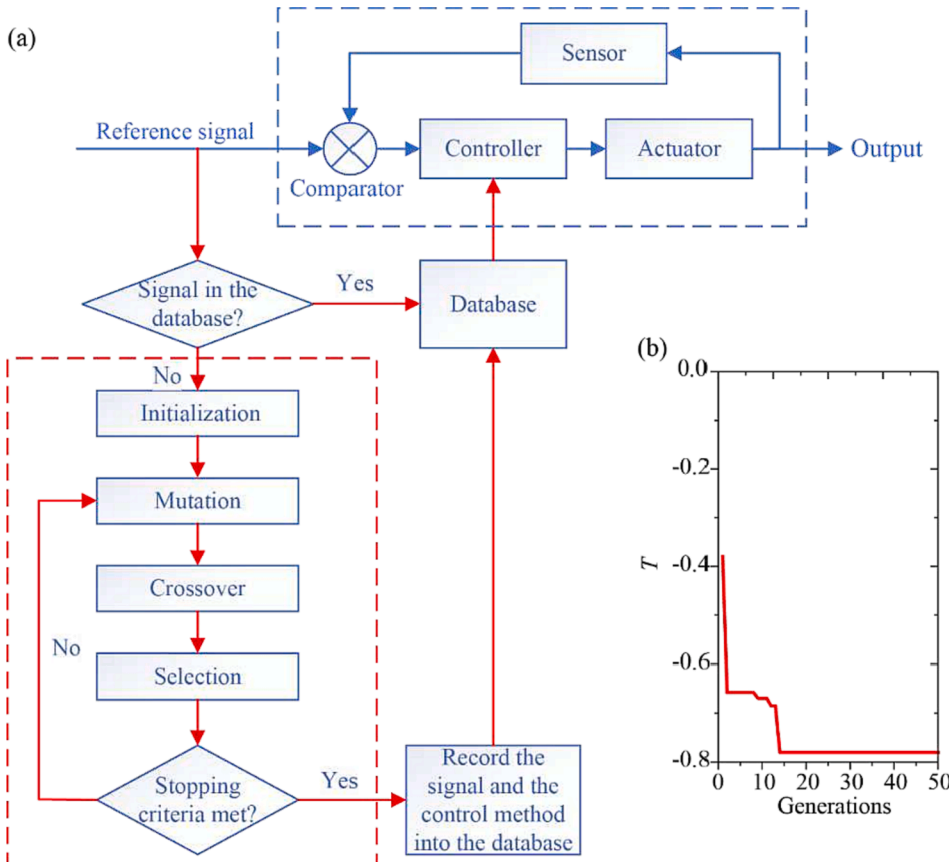


Fig. 13. (a) Flow chart of the self-adaptive MMSS. The flow chart inside the blue dashed frame is the active displacement feedback control. The flow chart inside the red dashed frame is the differential evolution (DE) optimization algorithm. (b) Convergence of the transmission coefficient T of the self-adaptive MMSS with the shunted PZT patches at 621 Hz in the ΓM direction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The convergence of the DE optimization for the self-adaptive MMSS in the ΓM direction at 621 Hz is shown in Fig. 13b. It can be seen that the best fitness function is already reached at the 13th generation. In most cases, the DE algorithm can generate an acceptable and convergent control gain within 15 generations. For the computational efficiency and accuracy, the 30th generation is chosen for the termination.

Next, the T values or transmission spectra for the MMSS-1 and the self-adaptive MMSS in the ΓX direction are shown in Fig. 14a, respectively. The corresponding displacement control gains are given in Fig. 14b. It can be seen that the self-adaptive MMSS has a wider band-gap compared to that of the MMSS-1. It can be also observed that the resonance-BG and the 1st Bragg-BG of the self-adaptive MMSS are merged into a wider one. Moreover, the band-gaps obtained by the transmission spectra agree well with the hypothetical band structures of the corresponding unit-cell via the self-sensing and self-actuating as shown in Fig. 11. In a word, the vibration or flexural wave mitigation is considerably enhanced by the merged wider band-gap of the self-adaptive MMSS.

To illustrate the vibration or flexural wave mitigation and isolation phenomena, the displacement fields of the flexural wave propagation in the ΓX direction of the MMSS-1 and the self-adaptive MMSS at 201 Hz and 401 Hz are shown in Fig. 15. It can be seen that the flexural waves cannot propagate in the self-adaptive MMSS with the displacement feedback control, while they can propagate through the MMSS-1 at 201 Hz. At 401 Hz, the mitigation or attenuation of the flexural waves in the self-adaptive MMSS is largely enhanced compared to that in the MMSS-1.

The transmission spectra T of the flexural waves for the MMSS-1 and with the self-adaptivity in the ΓM direction are shown in Fig. 16a. The corresponding displacement control gains are given in Fig. 16b. The mostly remarkable feature of the self-adaptive MMSS in the ΓM direction is that the resonance-BG, the 1st Bragg-BG and the 2nd Bragg-BG are merged to one wider band-gap. Hence, the flexural wave mitigation or attenuation increases significantly because of the merged band-gap (151–1251 Hz). The displacement fields of the flexural wave propagation in the ΓM direction of the MMSS-1 and the self-adaptive MMSS at 401 Hz are given in Fig. 17a and 17b, respectively. However, the merged band-gap is different from that (1001–1041 Hz) of the hypothetical band structures in Fig. 11. This discrepancy can be explained by the displacement fields of the flexural wave propagation in the ΓM direction of the MMSS-1 and the self-adaptive MMSS at 1021 Hz as shown in Fig. 17c and 17d. As we have already mentioned before, the 3rd band and the 4th band correspond to the symmetric and asymmetric flexural wave modes and they are folded together at the M point (see Fig. 9).

Only the asymmetric mode can be excited by a line source at the upper face-sheet, as shown in Fig. 17c. Hence, the 3rd band of the hypothetical band structures appears as a deaf band in the ΓM direction.

In summary, the proposed self-adaptive MMSS can achieve a wide merged band-gap and consequently a broadband vibration mitigation and isolation property. The self-adaptive strategy via the self-sensing and self-actuating is quite simple to realize. Hence, both the Bragg scattering and the local resonance band-gaps can be widened. In a further way, these band-gaps can be merged together like that in the hypothetically predicted band structures. In addition, the flexural wave mitigation or attenuation inside the band-gaps can be also significantly enhanced. The flexural waves propagating in the ΓX and ΓM directions are both investigated. The results demonstrate that the self-adaptive MMSS can select the corresponding displacement feedback control gains automatically according to the different excitations and boundary conditions by itself.

5. Conclusions

In this paper, a novel self-sensing and self-actuating metamaterial sandwich structure named as self-adaptive MMSS is proposed. The proposed self-adaptive MMSS is composed of the periodic hollow-tube core lattice sandwich structure (HLSS) with local resonators and PZT patches. To lend the self-sensing and self-actuating proficiency to the self-adaptive MMSS, the PZT patches are shunted by the active displacement feedback control. The active displacement feedback control parameters are generated automatically by the DE optimization algorithm according to the external excitation. Representative examples confirm that the proposed self-adaptive MMSS has an outstanding performance for the low-frequency vibration mitigation and isolation. The main contributions of this paper can be summarized as follows.

Firstly, the band structures and eigenmodes of the MMSS-1 are analyzed in details. The results show that the proposed MMSS-1 exhibits both Bragg-BGs and resonance-BGs. The Bragg-BGs are induced by the periodically distributed thin-walled hollow tubes as the core of the MMSS-1, while the resonance-BGs are generated by the local resonators attached to the periodic hollow tubes. The key geometrical and material parameters are analyzed to achieve wide band-gaps of the MMSS-1.

Secondly, we propose to attach the PZT patches onto the top and bottom cover-plate surfaces and the local resonators of the MMSS-1, which act as the sensors and the actuators, respectively. Then, the PZT patches are shunted via the active displacement feedback control. The effects of the displacement feedback control gains on the band-gaps of the MMSS-3 are analyzed. Subsequently, the target band structures are

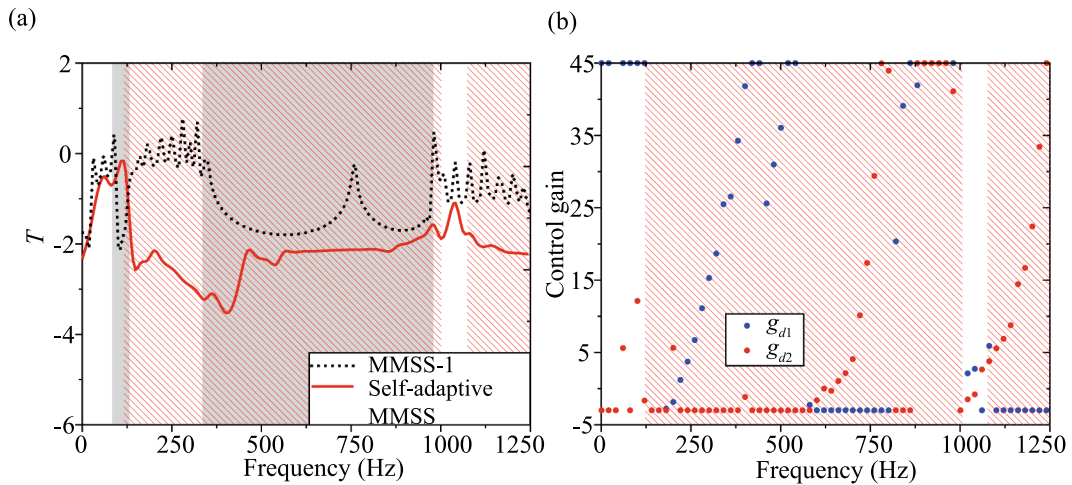


Fig. 14. (a) The transmission spectra (in ΓX direction) of the MMSS-1 and the self-adaptive MMSS. (b) The corresponding displacement feedback control gains of the self-adaptive MMSS in the ΓX direction. The resonance-BG and the 1st Bragg-BG of the MMSS-1 are shaded by gray, while the merged band-gaps of the self-adaptive MMSS are shaded by red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

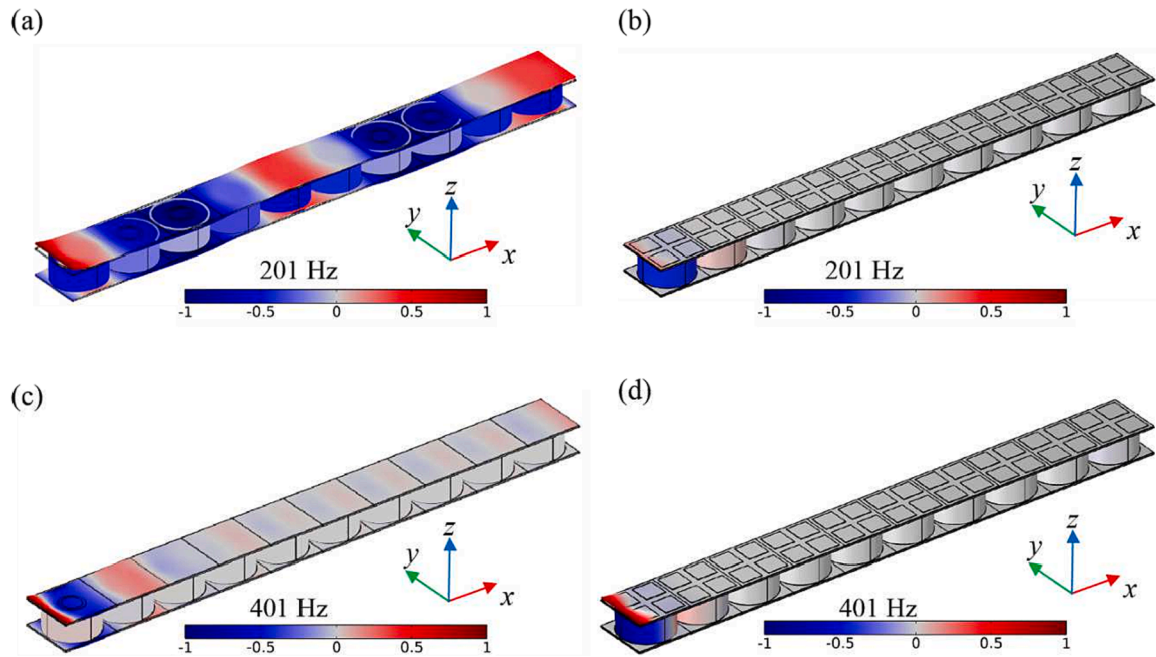


Fig. 15. The displacement fields of the flexural wave propagation (in ΓX direction) in the MMSS-1 at (a) 201 Hz and (c) 401 Hz, and in the self-adaptive MMSS at (b) 201 Hz and (d) 401 Hz.

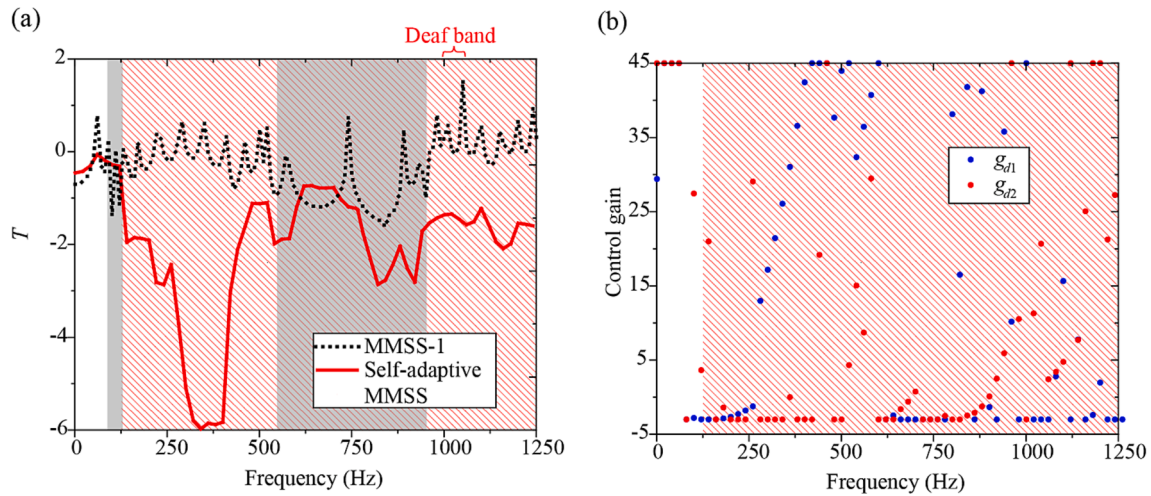


Fig. 16. (a) The transmission spectra (in ΓM direction) of the MMSS-1 and with the self-adaptive MMSS. (b) The corresponding displacement feedback control gains of the self-adaptive MMSS in the ΓM direction. The resonance-BG and the 1st Bragg-BG of the MMSS-1 are shaded by gray, while the merged band-gaps of the self-adaptive MMSS are shaded by red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

presented, which are used to identify the merged band-gaps of the self-adaptive MMSS.

Thirdly, the DE optimization algorithm is adopted to determine the best control parameters, which lend the self-sensing and self-actuating intelligence to the designed self-adaptive MMSS. Representative numerical simulation examples demonstrate that the proposed self-adaptive MMSS can widen both the Bragg-BG and the resonance-BG, and even merge them together into a much wider band-gap in the low-frequency range, which is essential for the broadband and substantial vibration mitigation and isolation.

The proposed self-adaptive MMSS may pave a novel and feasible way for the design and applications of the SSs for the low-frequency vibration mitigation and isolation in civil engineering, mechanical engineering, and aerospace engineering etc. Moreover, the other primary advantages of the SSs can be largely maintained since the proposed self-adaptive

strategy only barely changes their original structure. Finally, it should be mentioned that the proposed self-adaptive strategy via self-sensing and self-actuating can be also applied to other types of the SSs for improving their vibration mitigation and isolation performance, and for the NVH (noise, vibration and harshness) of vehicles, high-end equipment and high-precision instruments.

CRediT authorship contribution statement

Zheng-Yang Li: Conceptualization, Methodology, Software, Writing – original draft. **Yan-Zheng Wang:** Writing – original draft, Formal analysis. **Tian-Xue Ma:** Formal analysis. **Yong-Feng Zheng:** Formal analysis. **Chuanzeng Zhang:** Conceptualization, Funding acquisition, Project administration, Supervision. **Feng-Ming Li:** Conceptualization, Funding acquisition.

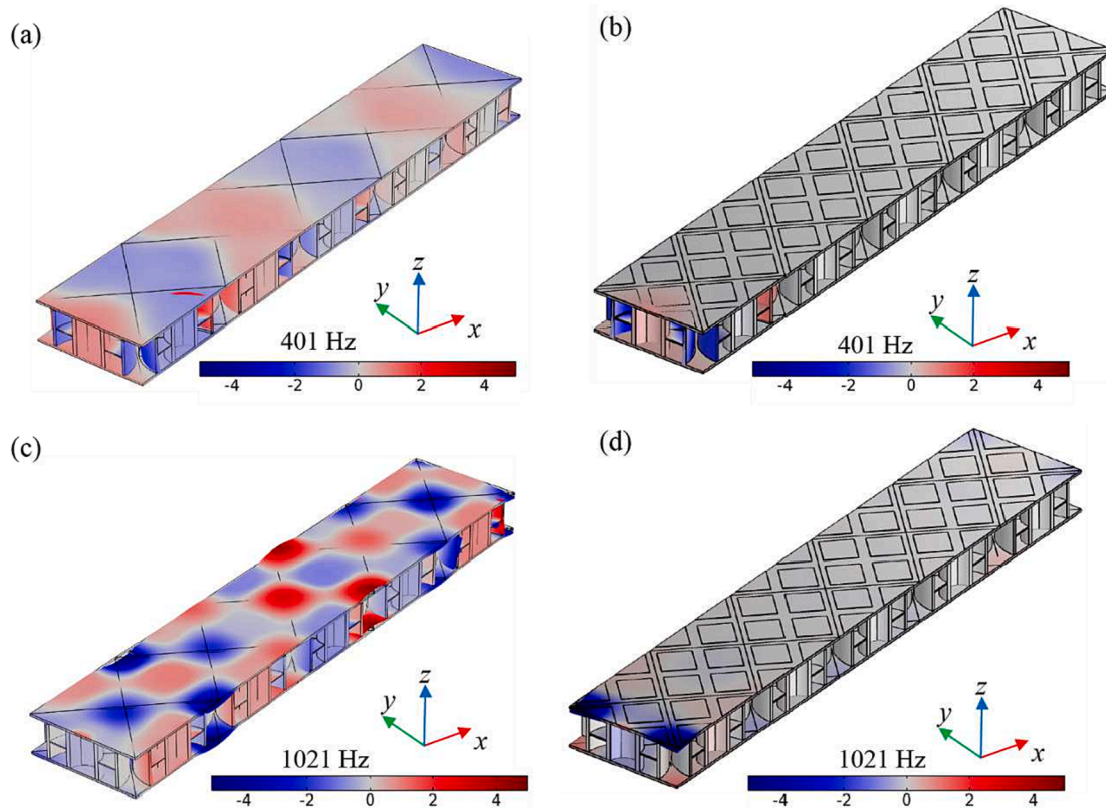


Fig. 17. The displacement fields of the flexural wave propagation (in ΓM direction) in the MMSS-1 at (a) 401 Hz and (c) 1021 Hz, and in the self-adaptive MMSS at (b) 401 Hz and (d) 1021 Hz.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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